ECP 8503: Intermediate Population PK/PD

Introduction to Pharmacodynamic Modeling

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17 April 2026

Outline

- Common PD structural equations
- Modeling delays
- Forcing functions

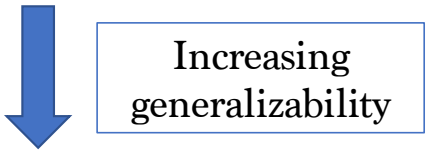
Pharmacokinetics is what the body does to the drug,
Pharmacodynamics is what the drug does to the body. . .

What does PK-PD modeling tell us?

- Quantify drug effect (PD) as a function of dose or drug exposure (PK)

- Can be structured to accommodate varying levels of information
  - E.g. for an anti-diabetic drug, do we have FPG measures, HbA1c measures, or both?

- Generalizability depends on information content
  - Classical approach (Empirical/direct/indirect)
  - Semi-mechanistic (TMDD/lifespan)
  - Mechanistic (QSP/PBPK)



- This lecture will focus primarily on empirical, direct, and indirect models

How do drugs affect the body?

- Typically, the concentration <u>at the site of action</u> directly correlates with an effect
  - Biological function is directly related to drug exposure
  - However, it is not trivial to sample the *effect site* for all disease states

- Depending on the type of endpoint, there may be a delay in the effect

Scales of Pharmacodynamic Endpoints

- Biomarker
  - CGRP (Migraine)
  - % CD4+ T cells (HIV)
- Imaging
  - Plaque density (Parkinson's)
  - Tumor size (solid tumors)
- Clinical Measures
  - HAM-D (Depression)
  - 6-Minute Walk Test (Muscular Dystrophy)

the molecule being targeted by a drug will typically show the fastest response

downstream effects such as structural changes typically follow a longer timescale

various factors impact clinical outcomes, so these changes may be even further delayed

Direct Model Relationships

- Equations can vary from simple to complex
  - Linear relationship
  - Piecewise linear splines
  - Exponential
  - Emax

- We are <u>almost always</u> simplifying biological processes
  - Capturing general trends using some mathematical formula
  - Depending on quality/amount of data, parameters may not be interpretable as a value that we can directly measure
  - All models should be “fit-for-purpose”

Empirical Models of Drug Effect

- Linear
  - Simplest
  - May be only option if limited observations
- Linear-exponential
  - Coefficient *m* allows for some curvature in E-R relationship
- Log-linear
  - May be useful with observations in 20-80% maximum effect range[1]
- Exponential
  - Decay processes

$$y = E_0 + S * Cp$$

$$y = E_0 + S * Cp^m$$

$$y = E_0 + m * \log(Cp)$$

$$y = E_0 * e^{-kt}$$

1: Ette. *Pharmacometrics*. 2007.

Linear Model NONMEM Code

```
$PK
;-----PK Model
CL = CLI
V  = VI
;-----PD Model
INTERCEPT = THETA(1)
SLOPE = THETA(2) * EXP(ETA(1))

$ERROR
EFFECT = INTERCEPT + SLOPE*F
Y = EFFECT + EPS(1) ; additive PD error
```

```
; example data file
; notice DV = measured EFFECT
$INPUT ID TIME  DV  AMT CLI  VI
;      1  0     4   .   3.5  6.1
;      1  0     .   25  3.5  6.1
;      1  0.5   12  .   3.5  6.1
;      1  2     52  .   3.5  6.1
;      1  6     60  .   3.5  6.1
```

Alternate approaches for including PK will be covered later in the lecture

Variables declared in $PK may be accessed in $ERROR but not vice versa

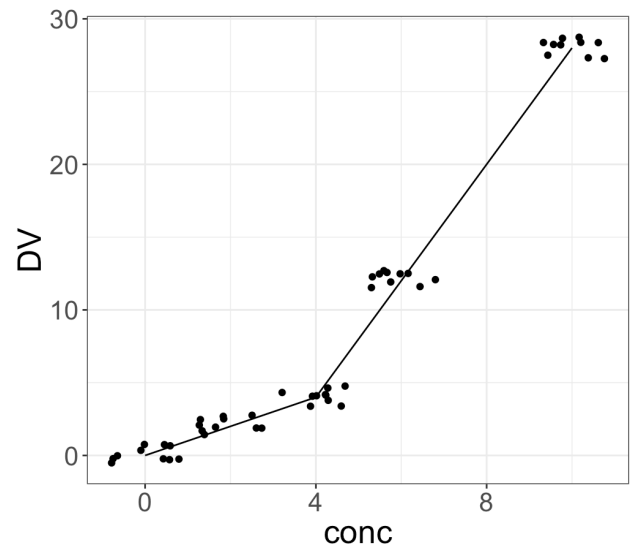
Linear Splines or "Hockey Stick" Model

- Compromise between linear and more complex models

$$y_{t1-t2} = m_1 x + b_1$$
$$y_{t2-t3} = m_2 x + b_2$$
$$y_{t3-\infty} = m_3 x + b_3$$

- Might be useful in situations where the entire dose-response relationship is not characterized

- It is helpful if you have some justification for splitting your data in this way



*NONMEM implementation included in backups

Abandon all hope, ye who extrapolate here

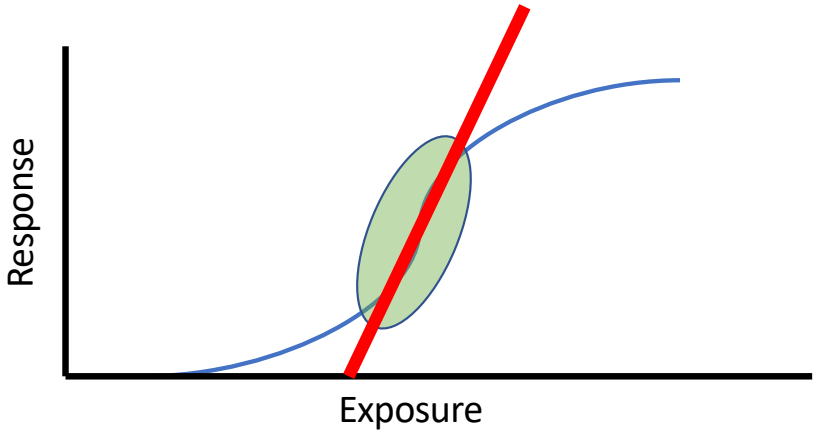
- Empirical linear or spline models can work well within a limited range of observed data

- However, you cannot know if this relationship exists beyond the observed data points

- Biological relationships are <u>rarely linear</u> across the entire exposure-response range (but they can be within defined sections)

The Gates of Hell. Auguste Rodin.
https://en.wikipedia.org/wiki/The_Gates_of_Hell

A typical exposure-response curve



- If your data fall within the linear phase, a linear model may be the only model that can be accurately estimated
- Studies should be designed to yield a broad range of exposures to get the best estimates of the exposure-response curve
- It is impossible to know when the curve will plateau without data supporting estimates of the full exposure-response curve

Emax Models

- Similar equation used in PK modeling to describe non-linear saturable processes (e.g. Michaelis-Menten binding)[1]

- Two estimated parameters
  - E_{max} = maximum effect
  - EC_{50} = concentration at 50% E_{max}

- At low concentrations ($Cp <<< EC_{50}$) approximates a linear relationship
  - $Slope = (E_{max} / EC_{50})$

$$E = \frac{E_{max} * Cp}{EC_{50} + Cp}$$

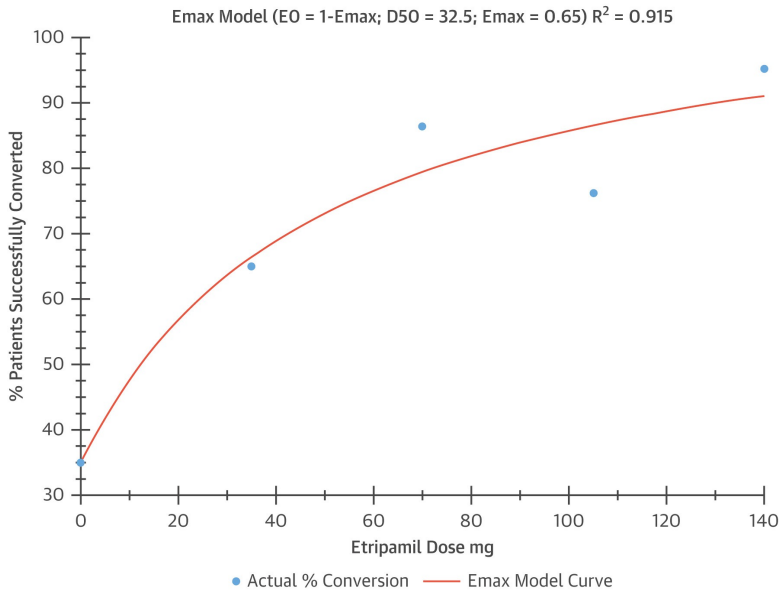


Image by Stambler et al. *J Am Coll Cardiol*. 2018 Jul 31;72(5):489-497. doi: 10.1016/j.jacc.2018.04.082.

1. Rowland & Tozer. *Clinical Pharmacokinetics and Pharmacodynamics 4e*., 2011

Sigmoidal Emax models

NO GAMMA ON MAXIMUM

$$E = \frac{E_{max} * Cp^{\gamma}}{EC_{50}{}^{\gamma} + Cp^{\gamma}}$$

- Derived from kinetics of oxygen binding to hemoglobin where it was used to predict the number of binding positions[1]
- The Hill Coefficient (γ) is a steepness factor
  - Increased γ allows for sigmoid or "S-shaped" curvature and a steeper increase in linear phase of the curve
  - Sigmoidicity allows a gradual increase in slope at low concentrations (pre-linear phase)
  - Collapses to Emax model at γ = 1

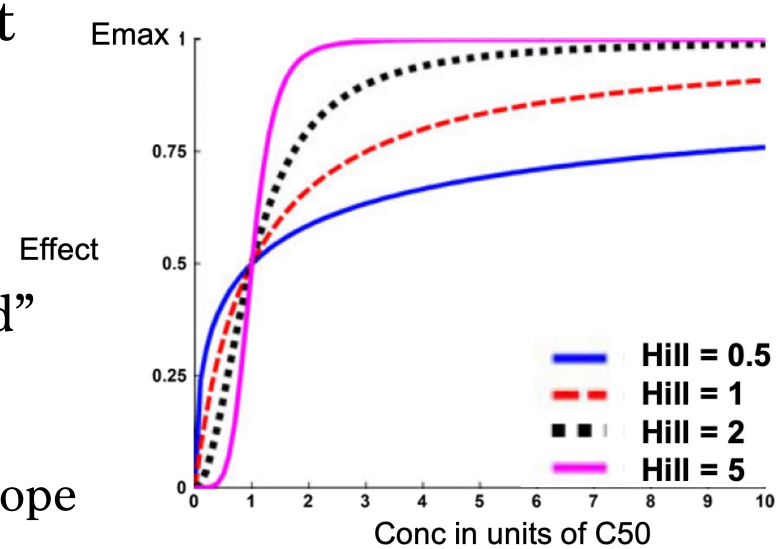


Image by Nick Holford
http://holford.fmhs.auckland.ac.nz/docs/immediate-time-course-of-drug-effect.pdf

1. Hill, AV. *J. Physiol.* 40 (1910): 4-7.

Sigmoid Emax Model NONMEM Code

```
$PK
;-----PK Model
CL = CLI
V  = VI
;-----PD Model
EMAX = THETA(1) * EXP(ETA(1))
EC50 = THETA(2)
GAMMA = THETA(3)

$ERROR
EFFECT = (EMAX * F**GAMMA)/(EC50**GAMMA + F**GAMMA)
Y = EFFECT + EPS(1) ; additive PD error
```

```
; example data file
; notice DV = measured EFFECT
$INPUT ID TIME  DV  AMT CLI  VI
;      1  0     4   .   3.5  6.1
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;      1  0.5   12  .   3.5  6.1
;      1  2     52  .   3.5  6.1
;      1  6     60  .   3.5  6.1
```
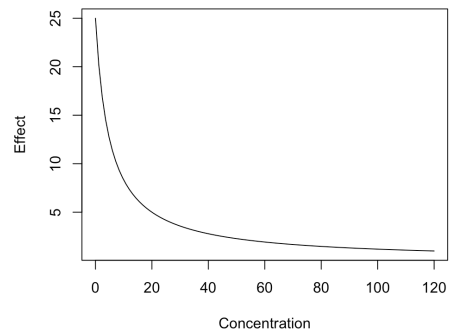

Note that Emax and EC50 are likely to be highly correlated; therefore, you will have trouble estimating an ETA on both parameters. Use your best judgment on which one gets IIV. . .

Extending Emax Models

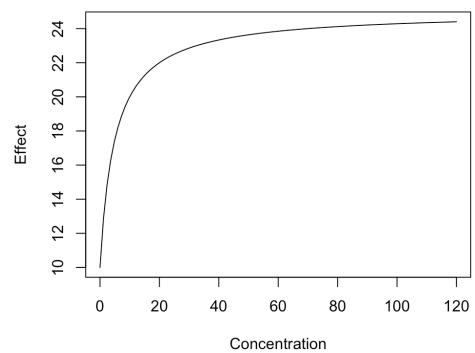
$$E = E_0 - \frac{E_{max} * Cp}{EC_{50} + Cp}$$



- Emin
  - Effect approaches a minimum
  - Baseline (E_0) may or may not be estimated

- Non-zero Baseline
  - Useful when modeling effect of drugs on endogenous compounds
  - Subtracting baseline from Emax makes drug effect depend upon <u>amount above baseline</u>
    - not always necessary
  - These constraints may not work when placing IIV on PD parameters

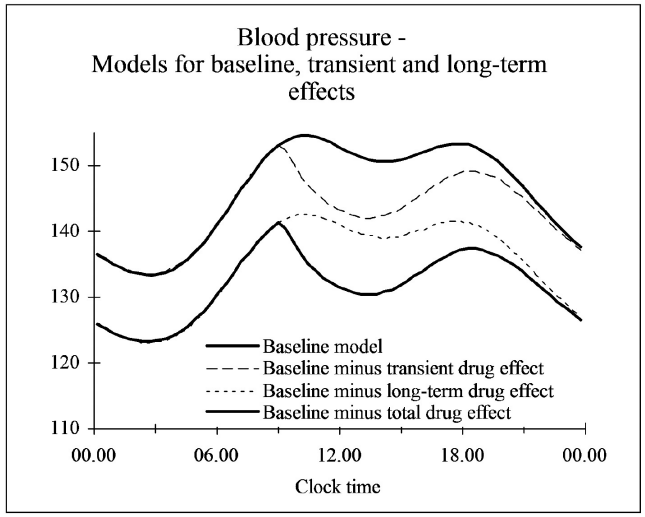
$$E = Base + \frac{(E_{max} - Base) * Cp}{EC_{50} + Cp}$$



Some approaches to incorporating baseline

$$E = E_0 + Emax * \frac{Cp}{EC_{50} + Cp}$$

- Estimation of the typical value and IIV of the baseline in the population ("Gold Standard")
- Incorporation of the baseline as a covariate while acknowledging measurement error
  - e.g. E0 comes directly from the data set
- Normalize the drug effect to baseline <u>in data set</u>
  - e.g. DVN = DV / BASE
- E0 may also be a function, like when baseline varies as a function of circadian rhythm
  - Blood pressure, cortisol, tQT



Hempel et al. *CPT.* 64(6):622-35;1998.

Modeling Delays with Indirect PK-PD Models

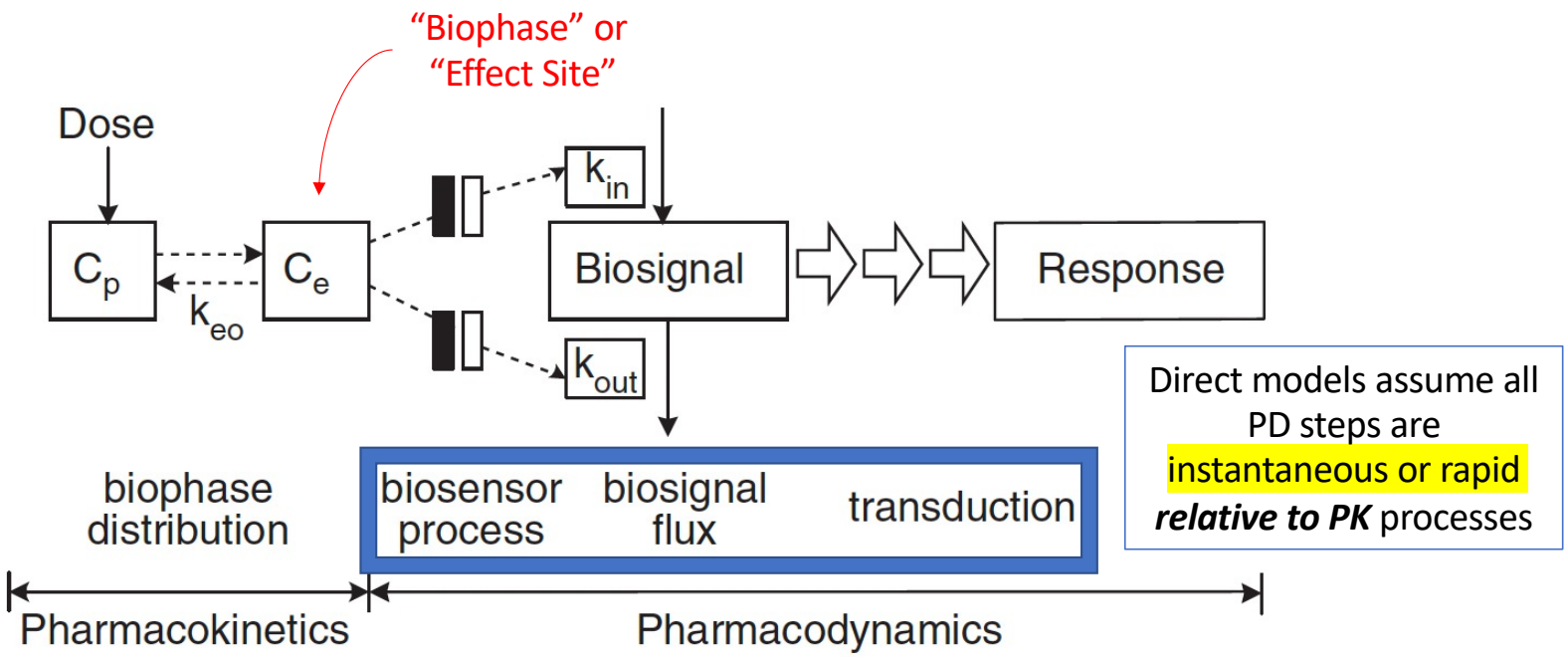


FIGURE 20.2 The different processes within pharmacodynamics. The biophase kinetics account for distributional kinetics between the plasma and effect site. The biosensor kinetics describe the dynamic behavior between drug and biosensing machinery. The biosignal is the first messenger in the transduction process that converts the signal to secondary messengers or the observed response. (Adapted from Ref. 19.)

Source: Ette, 2007, Chapter 20 (pg 530)

Graphical Detection of Delays

- Concentration & effect vs. Time

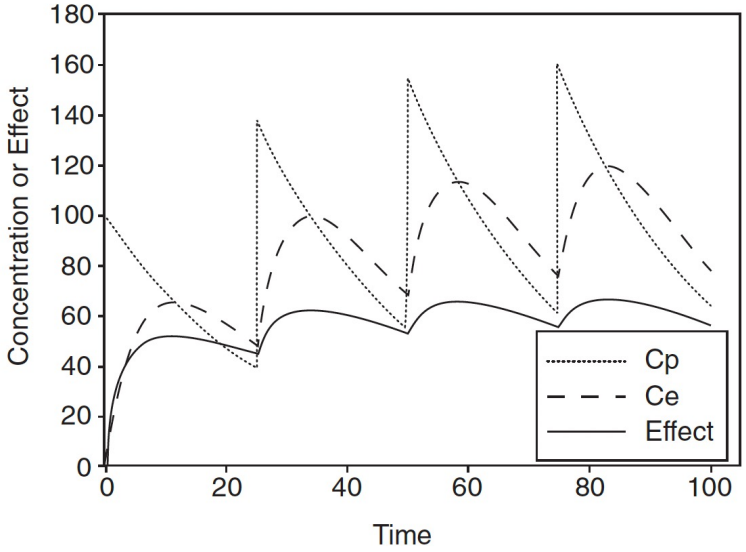


Figure 20.6; Ette, p535

Graphical Detection of Delays

- Concentration & effect vs. Time
- Hysteresis plots (effect vs. concentration)
  - Time delay
  - Changes in PD
  - Issues with assay/endpoint
  - Indirect relationship
  - Counter-clockwise = effect increases with time (common case)
  - Clockwise hysteresis = effect decreases with time ("tolerance")

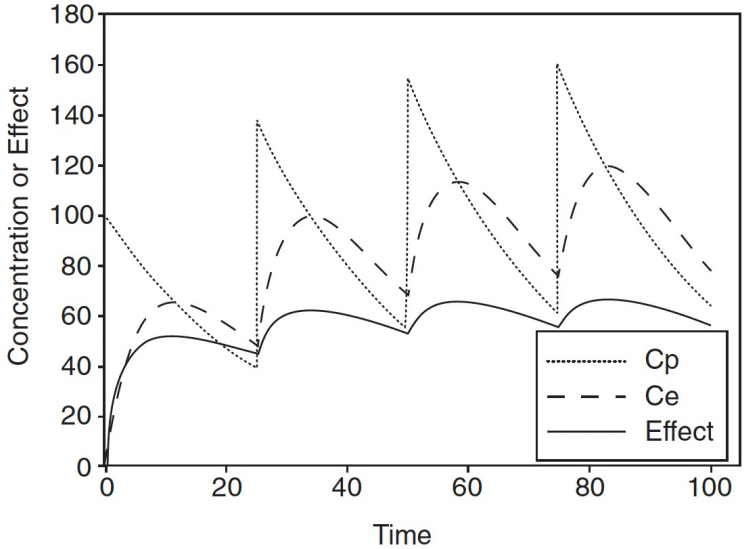
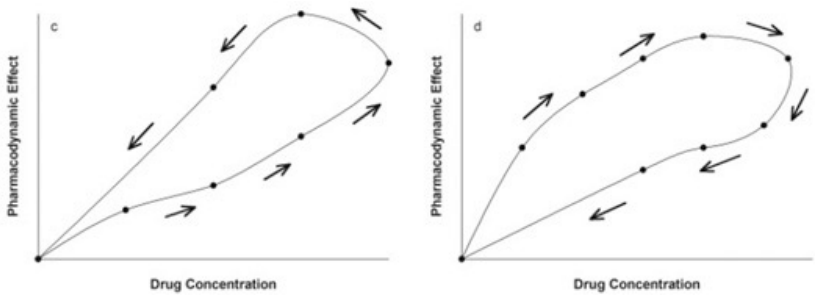


Figure 20.6; Ette, p535



Arrows indicate progression of time

Image adapted from Louizos et al. *J Pharm Pharm Sci*. 2014;17(1):34–91.

Graphical Detection of Delays

- Concentration & effect vs. Time
- Hysteresis plots (effect vs. concentration)
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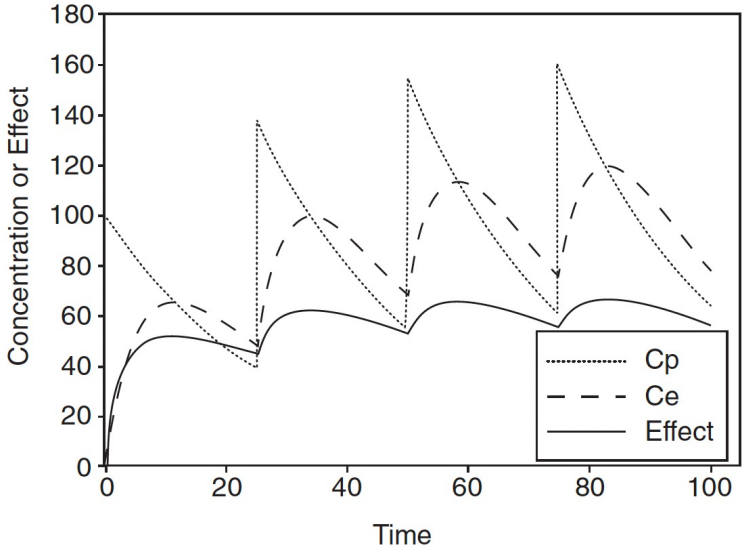
  - Clockwise hysteresis = effect decreases with time ("tolerance")
- <u>**Not always clear from plots alone**</u>



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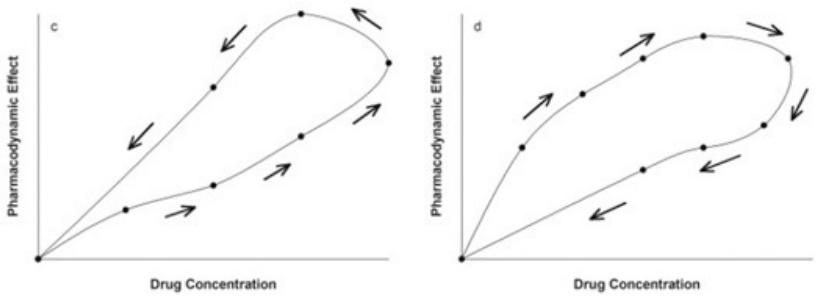


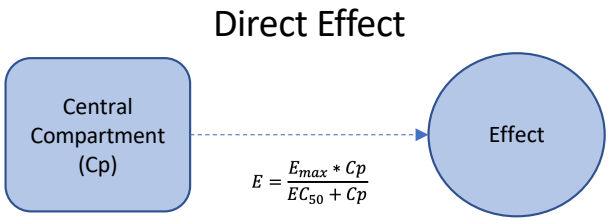
Arrows indicate progression of time

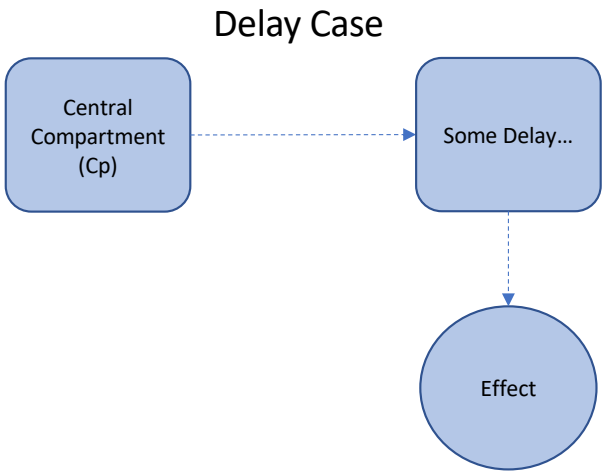
Image adapted from Louizos et al. *J Pharm Pharm Sci*. 2014;17(1):34–91.

Modeling Time Delays

- Effect Compartment
- Indirect Response Model
- Transduction/Lifespan Model
- Cell Binding Model

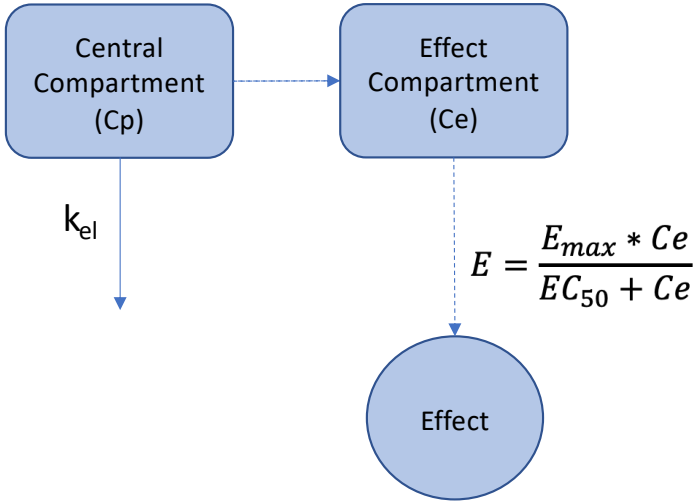
- Several others exist, the sky's the limit!





Effect Compartment

- Mass collected in a hypothetical effect compartment ("biophase") drives observed drug effect
- Delay occurs due to the added time for mass to travel into effect compartment
- Effect compartment does not necessarily represent biological partition



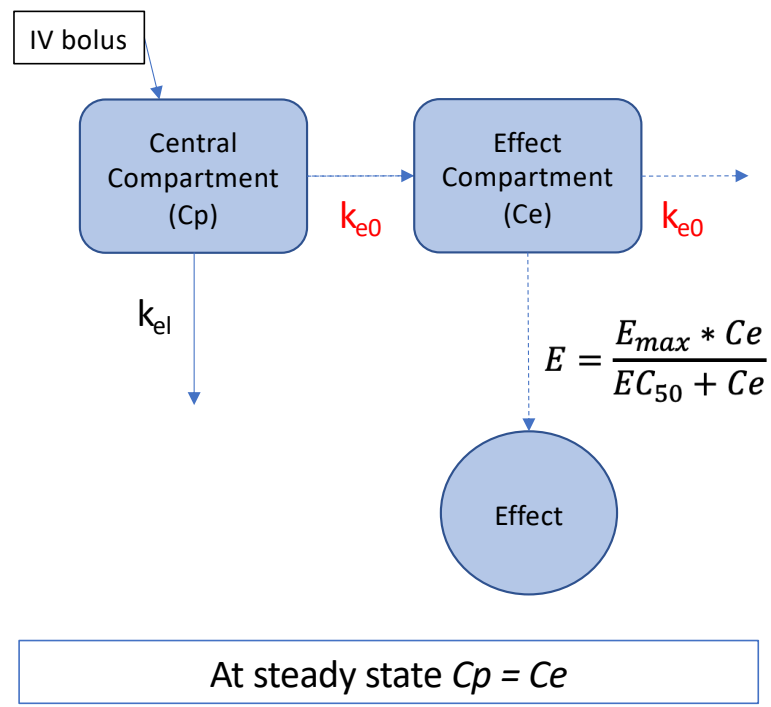
How does hypothetical mass accumulate?

"The effect compartment is modeled as an additional compartment linked to the plasma compartment by a first-order process, but whose exponential does not enter into the pharmacokinetic solution for the mass of drug in the body"

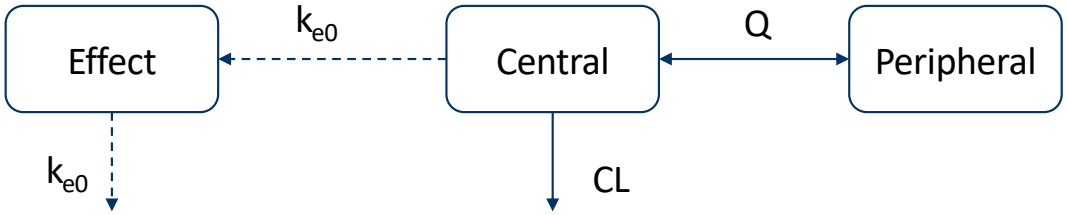
- Sheiner et al., 1979

$$\frac{dCp}{dt} = -kel * Cp$$

$$\frac{dCe}{dt} = ke_0 * Cp - ke_0 * Ce$$



Two-Compartment Implementation



- Don't want to perturb the existing PK model structure
- Effect compartment mirrors concentration in <u>central</u> compartment

Effect Compartment NONMEM Implementation

```
$SUBROUTINE ADVAN13
$MODEL        COMP (CENTRAL)
              COMP (EFFECT)
$PK
CL = CLI
V = VI
K10 = CL/V
KE0 = THETA(1)
EMAX = THETA(2)
EC50 = THETA(3) * EXP(ETA(1))

$DES
DADT(1) = -K10 * A(1)
DADT(2) =  KE0 * A(1)/V - KE0 * A(2)

$ERROR
CE = A(2)
EFFECT = EMAX*CE/(CE+EC50)
Y = EFFECT + EPS(1)
```

Notice A(2) is concentration, while A(1) is mass

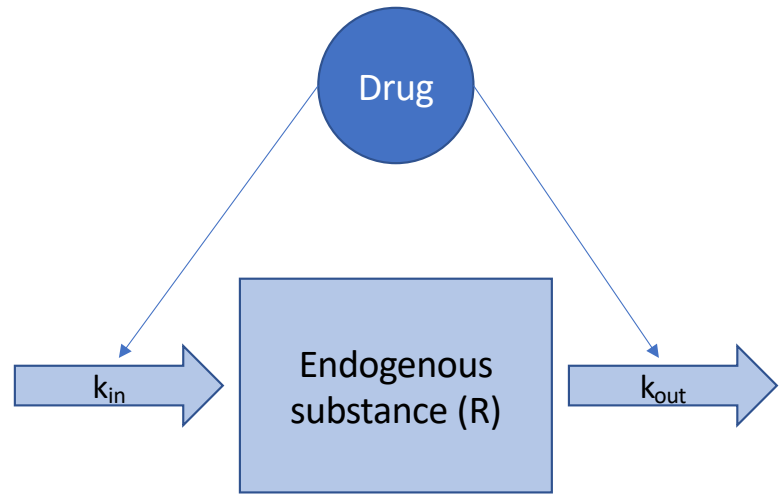
```
; example data file
$INPUT ID TIME  DV  EVID AMT  CLI  VI
;      1  0     4   0    .    3.5  6.1
;      1  0     .   1    10   3.5  6.1
;      1  0.5   12  0    .    3.5  6.1
;      1  2     52  0    .    3.5  6.1
;      1  6     60  0    .    3.5  6.1
```

Indirect Response Model

- Modeling an endogenous substance with a basal turnover rate

- Drug can stimulate/inhibit synthesis
  - Zero-order k_{in}

- Alternatively, drug can stimulate/inhibit degradation
  - First-order k_{out}

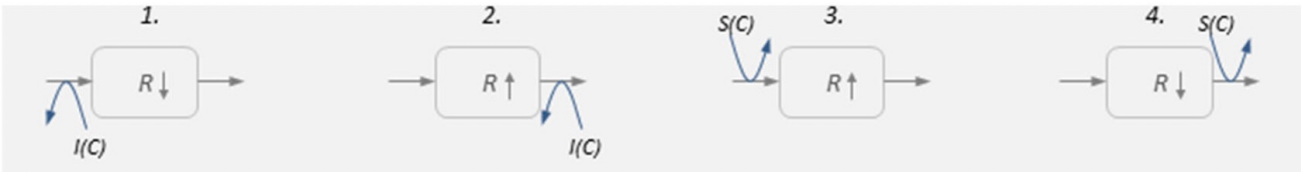


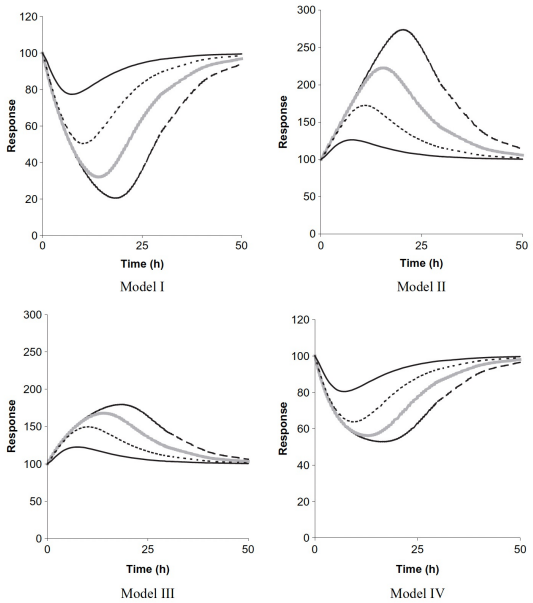
$$\frac{dR}{dt} = k_{in} - R * k_{out}$$

$$R_0 = \frac{k_{in}}{k_{out}}$$

Only need two of these parameters estimated

The Four Flavors of Indirect Response





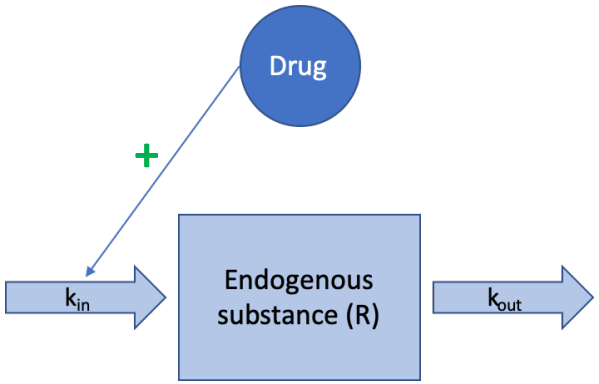
- Increasing effect with exposure
  - Stimulation of k_{in} (3)
  - Inhibition of k_{out} (2)
- Decreasing effect with exposure
  - Inhibition of k_{in} (1)
  - Stimulation of k_{out} (4)
- Mechanism should guide which option is chosen
- If mechanism unknown, study design may inform model choice

Ette. *Pharmacometrics*. 2007.
Gabrielsson J, Hjorth S. *AAPS J*. 2016;18(1):64-91.
Sharma A, Juska Wj. *Br J Clin Pharmacol.* 1998;45:229–239

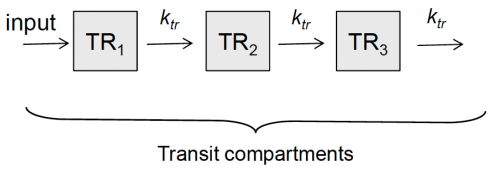
Stimulating k_{in} NONMEM Implementation

```
$SUBROUTINE ADVAN13
$MODEL  COMP (CENTRAL, DEFDOSE)
        COMP (RESPONSE)
$PK
; PK model
CL = CLI
V = VI
K10 = CL/V
S1=V

; Turnover model
E0 = THETA(1)
KOUT = THETA(2)
KIN = E0*KOUT
A_0(2) = E0

; Drug effect model
EMAX = THETA(4) * EXP(ETA(1))
EC50 = THETA(5)

$DES
CP = A(1)/V
DADT(1) = -K10*A(1)
DADT(2) = KIN*(1 + (EMAX*CP)/(CP+EC50)) - KOUT*A(2)
```

```
; example data file
$INPUT ID TIME  DV  CMT  AMT  CLI  VI
;      1  0     4   2    .    3.5  6.1
;      1  0     .   1    10   3.5  6.1
;      1  0.5   12  2    .    3.5  6.1
;      1  2     52  2    .    3.5  6.1
;      1  6     60  2    .    3.5  6.1
```



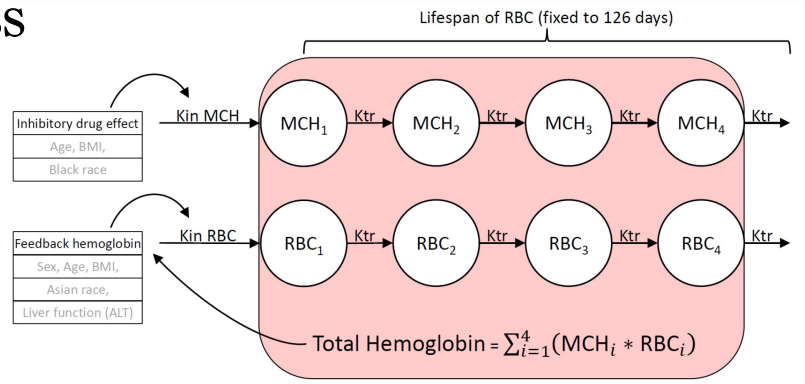
Transduction Model



$MTT=n/k_{tr}$

n= number of transitions
k_{tr}= first order rate constant between transit compartments
MTT= mean transit time

- "Friberg Model" – colloquially named after Lena Friberg (Uppsala)
  - *Friberg et al. Eur J Clin Pharmacol 2000*
- Discrete compartments may represent chains of reactions or different stages of a process
- The data may inform the number of compartments that are used

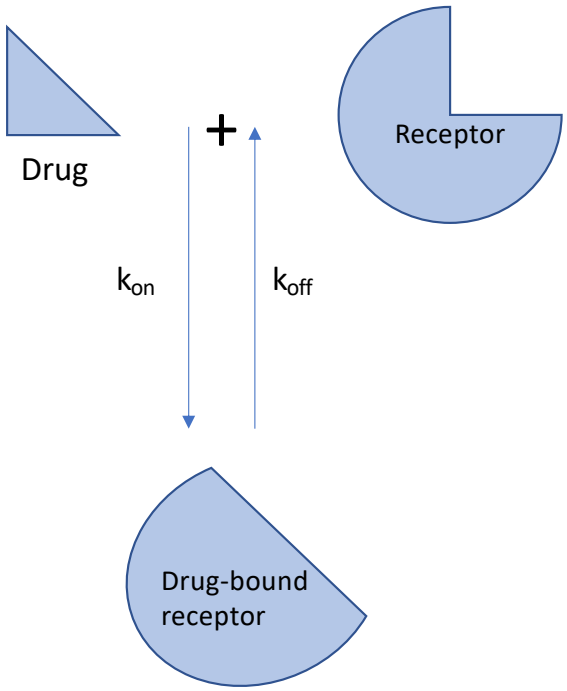


Hanke et al. PAGE 32 (2024) Abstr 10908

Binding Model

- Drug may either activate or suppress an effect through binding a receptor
  - Agonists activate
  - Antagonists inactivate

- Dissociation constant ($K_d = \frac{k_{off}}{k_{on}}$) is the concentration when half-maximal effect is reached
  - Analogous to EC_{50}



Okay, but what about the “PK” in “PK-PD”?

Bringing Exposures Into the Model

- Two alternative methods are most used to pull PK into PK-PD models
  - Simultaneous PK & PD modeling
  - Sequential PK & PD modeling

- Concentrations can also be pulled directly from data set
  - Time-matched exposures can be included as a column
  - Summary exposures may be used for responses with delayed effects that do not track concentration closely in time

Sequential vs. Simultaneous

- Two main approaches are used to link PK & PD data

- Simultaneously model PK & PD data
  - Still considered by many to be the ”gold standard”
  - However, many scientists continue to use sequential approach especially during model development to reduce runtimes

- PK & PD models combined using “forcing function”
  - Model PK then PD sequentially
  - Use PK output as input to PD model, estimate PD model parameters

Simultaneous PK-PD Modeling

- Including both data types allows the PD data to influence the estimates of PK parameters (and vice versa)
  - May be important if the drug effect has impact on drug PK
- Reliable estimates of uncertainty and precision of both PK & PD parameters
- Computationally burdensome and time consuming
- Model instability may result from trying to estimate too many parameters

Simultaneous NONMEM Implementation

```
$SUBROUTINE ADVAN1 TRANS2
$PK
 CL = THETA(1)*(WT/70)**0.75*EXP(ETA(1))
 V  = THETA(2)*(WT/70)**1.00*EXP(ETA(2))
 S1    = V
 BASE  = THETA(3)*EXP(ETA(3))
 SLOPE = THETA(4)

$ERROR
 EFF   = BASE + SLOPE*F

 Y     = F*EXP(EPS(1))
IF(DVID.EQ.2) Y = EFF+EPS(2)

$THETA
(0,10) ;1 CL
(0,100);2 V
(0,1)  ;3 BASE
(0,9)  ;4 SLOPE
```

Estimated drug concentration drives effect

```
; example data file
$INPUT ID TIME  DV  AMT  DVID
;      1  0     12  .    2    ;PD
;      1  0     .   10   0    ;DOSE
;      1  2     27  .    1    ;PK
;      1  2     8   .    2    ;PD
```

DVID flag used to determine if PK or PD modeled

Sequential Modeling Options

- Import Individual PK Parameters from PK Model
  - Individual PK Parameters (**IPP** – common case)
    - Assume PK model estimates are without error
  - Individual PK Parameters and Standard Error (**IPPSE**)
    - Includes uncertainty around PK estimates
- Import Population PK Parameters from PK Model
  - Population PK Parameters (**PPP**)
    - Inter-individual variability comes from PD observations
  - Population PK Parameters and PK Data (**PPP&D**)
    - Include PK data for inter-individual variability estimates

Zhang et al. J Pharmacokinet Pharmacodyn. 2003 Dec;30(6):387-404.
Lacroix et al. J Pharmacokinet Pharmacodyn. 2012 Apr;39(2):177-93.

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Individual PK Parameters (IPP)

- PK model fit to data <u>entirely separate from</u> PD data
  - Estimates of PK parameters included in PD dataset
  - PK data not included in PD dataset

- Typically results in a more stable model than simultaneous modeling approach

- May introduce a bias especially in subjects with sparse PK sampling
  - For subjects with few observations, EBEs will shrink toward population average
  - Estimates of PD variability downwardly biased because uncertainty in PK parameter estimates is ignored

IPP NONMEM Implementation

Can use same data set if you add `IGNORE(DVID.EQ.1)` in `$DATA`

```
$SUBROUTINE ADVAN13
$MODEL      COMP (CENTRAL, DEFDOSE)
            COMP (EFFECT)
$PK
CL = CLI
V = VI
K10 = CL/V
KE0 = THETA(1)

$DES
DADT(1) = -K10 * A(1)
DADT(2) = KE0 * A(1)/V - KE0 * A(2)

$ERROR
CE = A(2)
EMAX = THETA(2)
EC50 = THETA(3) * EXP(ETA(1))
EFFECT = EMAX*CE/(CE+EC50)
```

```
; example data file
; DV is a measure of PD
$INPUT ID TIME  DV  DVID AMT CLI  VI
;      1  0     .   0    10  3.5  6.1
;      1  0.5   4   2    .   3.5  6.1
;      1  1.5   12  2    .   3.5  6.1
;      1  2     18  2    .   3.5  6.1
;      2  0     .   0    10  3.2  6.3
```

Note that the individual parameter estimates already contain covariate effects & IIV

Population Pharmacokinetic Parameters & Data (PPP&D)

- Data set needs to include PK & PD data for all individuals
  - Accounts for variability in PK parameter precision
  - Estimates of ETAs on PD parameters are unbiased using this approach
- Diagnosis of PK distortion from PD data possible with this approach, offering a distinct advantage over IPP methods
- PD parameter SE is underestimated, but improved from IPP methods

PPP&D NONMEM Implementation

```
$PK
CL = THETA(1)*(WT/70)**0.75*EXP(ETA(1))
V  = THETA(2)*(WT/70)**1.00*EXP(ETA(2))

$ERROR
BASE = THETA(3) * EXP(ETA(1))
SLOPE = THETA(4)
EFFECT = BASE + SLOPE*F
Y=F*(1+EPS(1))
IF(DVID.EQ.2) Y = EFF+EPS(2)

$THETA
3.5 FIX          ; 1-CL
6.1 FIX          ; 2-V
(0,1)            ; 3-BASE
(0,5)            ; 4-SLOPE

$OMEGA
0.1 FIX          ; IIV CL
0.1 FIX          ; IIV V
0.04             ; IIV BASE
0.04             ; IIV SLOPE

$SIGMA
0.2 FIX          ; RUV PK
0.1              ; RUV PD
```

The PK model here <u>does</u> need to include any covariate effects and IIV

```
; example data file
$INPUT ID TIME  DV  AMT  DVID
;      1  0     .   10   0    ;DOSE
;      1  0.5   12  .    1    ;PK
;      1  2     22  .    1    ;PK
;      1  2     0.6 .    2    ;PD
```

All PK parameters are fixed to estimates from PK model, even random effects, otherwise <u>identical</u> to SIM control stream

Comparing Simultaneous & Sequential

Table II. Estimation-status Statistics ($N = 200$)

		SIM	PPP	PPP&D	IPP
I:	% of studies that converged	100.0	100.0	100.0	100.0
II:	% of studies among I for which all variance components could be estimated	40.0	42.5	45.5	59.0
III:	% of studies among II for which SE could be estimated	95.0	96.4	98.9	100.0

Zhang et al. J Pharmacokinet Pharmacodyn. 2003 Dec;30(6):387-404.

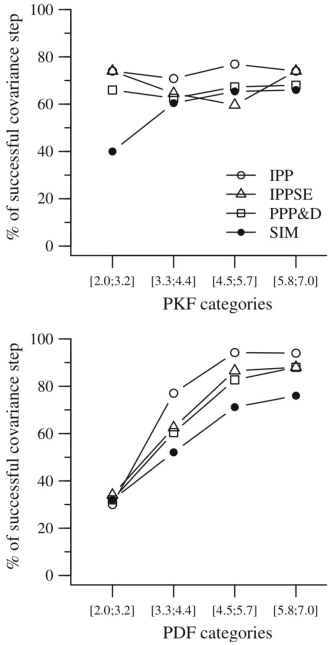


Fig. 6 Frequency of successful covariance step by mean number of PD and PK observations in the datasets (PKF and PDF). *Empty circles*, *squares* and *triangles* represent the sequential methods IPP, PPP&D and IPPSE, respectively and *filled squares* the SIM method

IPPSE performs similar to PPP&D

Lacroix et al. J Pharmacokinet Pharmacodyn. 2012 Apr;39(2):177-93.

Alternative Exposure Measures

- Rather than plasma concentration at a matched timepoint driving drug effect, summarized exposures can be used
- Measures of cumulative/summary exposure can include:
  - $AUC_{0\text{-}t}$, $C_{avg,ss}$, C_{max}, C_{min}
- When would this be necessary?
  - Clinical/biological hypothesis (e.g. AUC effect on FPG in T2DM[1])
  - Time-matched models not adequately capturing trends in data
- This approach is not appropriate in situations when the response acutely tracks plasma concentration

1. Baron KB et al., Diabetes Ther. 2016 Sep;7(3):455-71.

NONMEM Implementation: PK-PD Using AUC

```
$SUBROUTINE ADVAN13
$MODEL    COMP (RESPONSE)
$PK
; Turnover model
KIN = THETA(1)
KOUT = THETA(2)
A_0(1) = KIN/KOUT

; Drug effect model
EMAX = THETA(4)
EC50 = THETA(5)

$DES
DADT(1) = KIN - KOUT*A(1)*(1 + (EMAX*AUC)/(AUC+EC50))
```

```
; example data file
$INPUT ID TIME  DV  CMT  AMT  AUC
;      1  -0.5  4   1    .     0
;      1  0     12  1    .   250
;      1  2     52  1    .   250
;      1  6     60  1    .   250
```

Summary

- Several structural models can be used to relate PK to PD
  - Linear/Spline/Emax/Sigmoidal Emax
- PK can be directly linked to effect, or delays can be incorporated
- Several methods are available for incorporating delay
  - Effect compartment, indirect response, binding/transduction/lifespan
- PK can come directly from data set, modeled simultaneously with PD, or PK model results can be added to the data set
- The multitude of choices reflect the variety of relationships that can exist between exposures and response

Backups

Linear Spline Model NONMEM Code

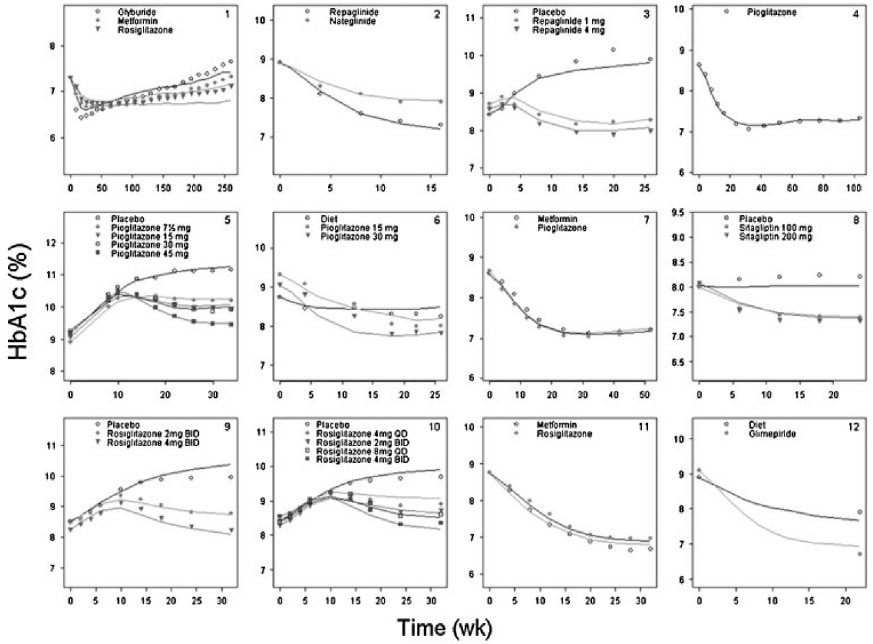
```
$PK
;-----PK Model
CP = CONC            ; CONC in dataset
;-----PD Model
INT1 = THETA(1)
SLOPE1 = THETA(2) * EXP(ETA(1))
INT2 = THETA(3)
SLOPE2 = THETA(4) * EXP(ETA(2))

IF(CP < 25)
EFFECT = INT1 + SLOPE1*CP
ELSE
EFFECT = INT2 + SLOPE2*CP

$ERROR
Y = EFFECT + EPS(1) ; additive PD error
```

```
; example data file
; notice DV = measured EFFECT
$INPUT ID  TIME  DV  AMT CONC
;      1   0     0   .   0
;      1   2     2   .   2
;      1   4     4   .   4
;      1   6     12  .   6
;      1   10    28  .   10
```

Adding a Physiological Lower Limit

$$\frac{dX}{dt} = kin - kout * X * (1 - \frac{X_{LL}}{X})$$

- Commonly applied in diabetes research (X = HbA1c)[1]
- k_{out} driving force is the *difference* between X and lower limit (X_{LL})
- May represent physiological limit or a clinician-induced lower limit (e.g. titrating dose down to 7% HbA1c)



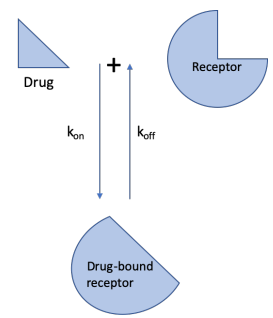
1: Samtani M. *Biopharm Drug Dispos*. 2010;31(3):135-143.

Antagonist Binding Model NONMEM Implementation

```
$SUBROUTINE ADVAN13
$MODEL  COMP (REC)
        COMP (BOUND_REC)
$PK
; PK model
CP = CONC

; PD model
KON = THETA(1)*EXP(ETA(1))
KOFF = THETA(2)*EXP(ETA(2))
KD = KOFF/KON
A_0(1) = 1

$DES
DADT(1) = -KON*CP*A(1) + KOFF*A(2)
DADT(2) =  KON*CP*A(1) - KOFF*A(2)

$ERROR
BASE = THETA(3)
IPRED = BASE * A(1)
```

Notice k_{on} is a second-order rate constant

```
; example data file
$INPUT ID TIME  DV  CONC
;      1  0     27  .
;      1  0.5   22  12
;      1  2     18  24
;      1  6     13  40
```



Special Case of Effect Compartment

- Closed-form solutions in NONMEM can be modified to convert one of the compartments to an effect compartment
  - ADVAN 3/4 for one-compartment + effect compartment
  - ADVAN 11/12 for two-compartment + effect compartment

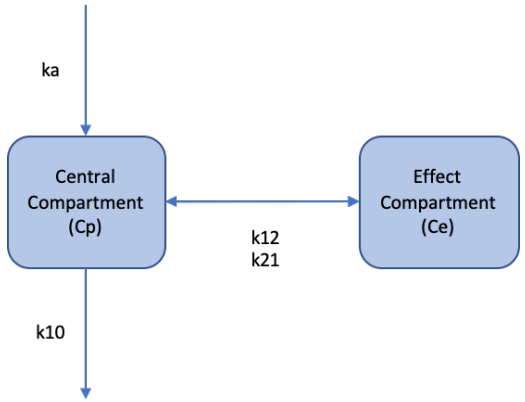
- ADVANs use closed-form solutions to differential equations
  - Faster runtimes
  - Beneficial for large datasets

- Adventurous modelers can write closed form equations in $PRED for a multitude of PK models
  - Colburn WA. J Pharmacokinet Biopharm. 1981 Jun;9(3):367-88.

Adjusting ADVAN3 for Effect Compartment

```
$SUBROUTINE ADVAN3 TRANS1

$PK
CL = THETA(1)*EXP(ETA(1))
V = THETA(2)*EXP(ETA(2))
K10 = CL/V
K12 = 0.001*K10 ; negligible loss of mass
K21 = THETA(3)
S1 = V
S2 = S1*K12/K21 ; preserves CeSS = CpSS

$ERROR
EMAX = THETA(4)
EC50  = THETA(5)*EXP(ETA(4))
IF(CMT.EQ.1)  Y  = F*EXP(EPS(1))
IF(CMT.EQ.2)  Y  = EMAX*F/(EC50+F)+ EPS(2)
```

```
; example data file
$INPUT ID  TIME  DV  CMT  AMT
;      1   0     .   1    10
;      1   0     4   2    .
;      1   0.5   12  2    .
;      1   2     3   1    .
;      1   2     52  2    .
;      1   6     60  2    .
```



Individual PK Parameters & Standard Errors (IPPSE)

- Improves upon IPP method by incorporating precision of parameter estimates into PK-PD model
- Acknowledges uncertainty in PK parameter estimate, but constrains estimation to the standard error estimate from PK model
- Circumvents underestimation of PD parameter uncertainty
- Decreased run time relative to simultaneous

IPPSE NONMEM Implementation

```
$PK
CL = CLI*EXP(ETA(1)*SECL)
V = VI*EXP(ETA(2)*SEV)

$ERROR
EMAX = THETA(1)
EC50 = THETA(2) * EXP(ETA(3))
EFFECT = EMAX*CE/(CE+EC50)
Y = EFFECT*(1+EPS(1))

$THETA
(0, 1)       ; 1-EMAX
(0, 18)      ; 2-EC50

$OMEGA
1 FIX ; IIV CL
1 FIX ; IIV V
0.04  ; IIV EC50
```

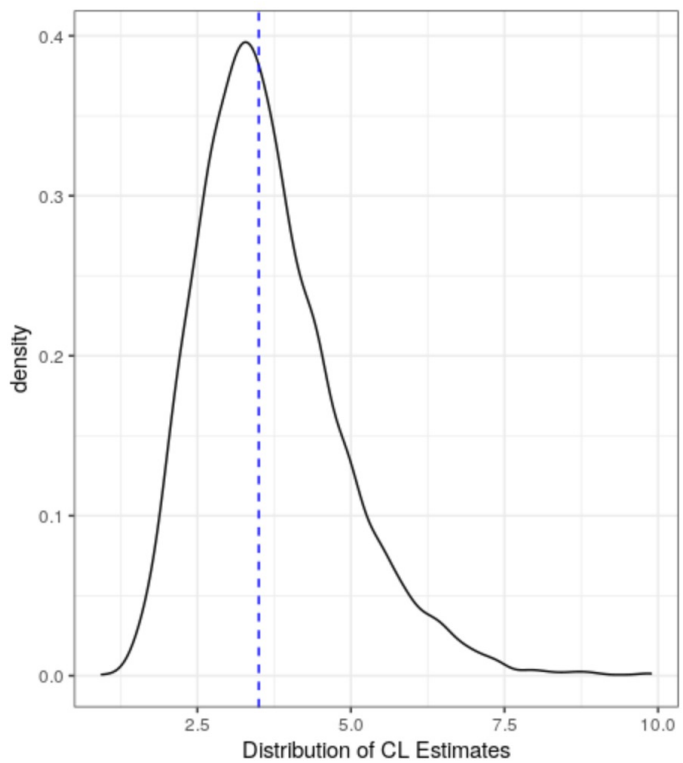
ETAs used for uncertainty on PK params in this model

```
; example data file
$INPUT ID TIME  DV  CMT  AMT  CLI  SECL  VI   SEV
;      1  0     .   1    10   3.5  0.09  6.1  0.11
;      1  0.5   12  2    .    3.5  0.09  6.1  0.11
;      1  2     22  2    .    3.5  0.09  6.1  0.11
;      2  0     .   1    10   3.2  0.08  6.3  0.13
```

SE(PARAM) imported from root.PHI file

Why is the initial estimate of ETA(1&2) 1 FIX?

```
$PK
CL = CLI*EXP(ETA(1)*SECL)
V = VI*EXP(ETA(2)*SEV)

$ERROR
EMAX = THETA(1)
EC50 = THETA(2) * EXP(ETA(3))
EFFECT = EMAX*CE/(CE+EC50)
Y = EFFECT*(1+EPS(1))

$THETA
(0, 1)       ; 1-EMAX
(0, 18)      ; 2-EC50

$OMEGA
1 FIX ; IIV CL
1 FIX ; IIV V
0.04  ; IIV EC50
```



CLI	SECL	VI	SEV
3.5	0.09	6.1	0.11
3.5	0.09	6.1	0.11
3.5	0.09	6.1	0.11
3.2	0.08	6.3	0.13

1 FIX = draw from log-normal distribution with SD = 1
1*0.09 FIX = draw from log-normal distribution with SD = 0.3

Population Pharmacokinetic Parameters (PPP)

- Only population-level estimates are imported from the pop PK model
  - Variability in PK parameter estimates is added through fixed ETAs

- Unlike IPP methods, PPP assumes all individuals exhibit the population estimate for their PK parameters

- Applicable in situation where PK data unavailable from similar population with PD data, otherwise to imprecise to consider for use
  - E.g. published pop PK model results

PPP NONMEM Implementation

```
$PK
CL = THETA(1)*EXP(ETA(1))
V = THETA(2)*EXP(ETA(2))

$ERROR
BASE = THETA(3) * EXP(ETA(1))
SLOPE = THETA(4)
EFFECT = BASE + SLOPE*F

$THETA
3.5 FIX       ; 1-CL
6.1 FIX       ; 2-V
(0,1)         ; 3-BASE
(0,5)         ; 4-SLOPE

$OMEGA
0.1 FIX       ; IIV CL
0.1 FIX       ; IIV V
0.04          ; IIV BASE
0.04          ; IIV SLOPE
```

```
; example data file
$INPUT ID TIME  DV  AMT
;      1  0     .   10
;      1  0.5   12  .
;      1  2     22  .
;      2  0     .   10
```